Project Report Format

1. INTRODUCTION

1.1 Project Overview

UCABS is a full-stack ride-hailing platform developed using the **MERN stack (MongoDB, Express.js, React.js, Node.js)**. The system connects passengers with nearby drivers through a digital platform that enables real-time ride booking and trip management.
The platform consists of three major modules:

- Rider application for booking rides
- Driver dashboard for managing trips
- Admin panel for driver verification and system monitoring

UCABS integrates **Socket.io for real-time ride tracking** and **TomTom Maps API for route navigation and distance calculation**. The system ensures security through **OTP-based ride verification** and user authentication.

1.2 Purpose

The purpose of this project is to design and implement a **smart transportation system** that simplifies ride booking for users and provides drivers with a platform to accept ride requests efficiently.
The project aims to:

- Provide fast and reliable ride booking
- Enable real-time GPS tracking of rides
- Ensure security through OTP verification
- Provide an admin system for driver KYC verification
- Deliver a user-friendly interface for passengers and drivers

2. IDEATION PHASE

2.1 Problem Statement

Traditional transportation systems often suffer from problems such as lack of driver availability, difficulty in tracking rides, and security concerns for passengers.

Users often face issues like:

- **Long waiting time for taxis**
- **No real-time tracking**
- **Lack of ride confirmation security**
- **Poor driver verification systems**

Therefore, there is a need for a digital ride-hailing platform that ensures convenience, safety, and efficiency for both passengers and drivers.

2.2 Empathy Map Canvas

User Thinks

- **I want a quick ride to my destination**
- **I need a safe driver**
- **I want to track my ride**

User Feels

- **Frustrated when rides are unavailable**
- **Unsafe when driver details are unknown**

User Says

- **I want a reliable ride booking system**
- **I want to know the fare before booking**

User Does

- **Searches for nearby taxis**
- **Uses ride booking applications**

2.3 Brainstorming

During brainstorming, several ideas were discussed to improve the ride booking experience. Possible ideas included:

- **Real-time ride tracking**
- **Driver verification system**
- **Fare estimation before booking**
- **OTP verification for ride start**
- **Admin control panel**
- **Secure login system**

The final solution chosen was a MERN-based ride-hailing platform with real-time tracking and secure ride verification.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

1. **User opens the application**
2. **User enters pickup and drop location**
3. **System calculates estimated fare**
4. **Ride request is sent to nearby drivers**
5. **Driver accepts the ride**
6. **Rider receives driver details**
7. **Rider verifies OTP before trip starts**
8. **Ride begins and GPS tracking starts**
9. **Ride completes and trip history is saved**

3.2 Solution Requirement

Functional Requirements

- **User registration and login**
- **Ride booking functionality**
- **Driver ride acceptance**
- **Real-time GPS tracking**
- **OTP ride verification**
- **Admin verification of drivers**

Non-Functional Requirements

- **Secure authentication**
- **Fast API response**
- **Scalable system architecture**
- **Reliable database management**

3.3 Data Flow Diagram

Level 0 DFD (System Overview)

User → Ride Request → System → Driver

Driver → Accept Ride → System → Rider

Admin → Verify Driver Documents → System

3.4 Technology Stack

Frontend

- **React.js**
- **HTML**
- **CSS**
- **JavaScript**

Backend

- **Node.js**
- **Express.js**

Database

- **MongoDB**

APIs & Tools

- **Socket.io (real-time communication)**
- **TomTom Maps API (navigation & routing)**
- **JWT Authentication**

4. PROJECT PLANNING & SCHEDULING

4.1 Problem Solution Fit

The UCABS platform solves the problem of inefficient ride booking by providing:

- real-time driver matching
- secure authentication
- transparent pricing
- reliable ride tracking

This ensures convenience for passengers and efficient ride management for drivers.

4.2 Proposed Solution

The proposed system is a **web-based ride-hailing application** built with MERN stack.

The system contains:

- Rider interface for booking rides
- Driver dashboard for accepting trips
- Admin panel for managing drivers and verifying documents

The system also integrates **real-time communication for ride updates** and **mapping services for navigation.**

4.3 Solution Architecture

The architecture follows a **three-layer structure**:

Frontend Layer
React.js user interface.

Backend Layer
Node.js and Express.js REST APIs.

Database Layer
MongoDB for storing user, driver, and ride data.

Communication occurs through **HTTP APIs and Socket.io events.**

5. FUNCTIONAL AND PERFORMANCE TESTING

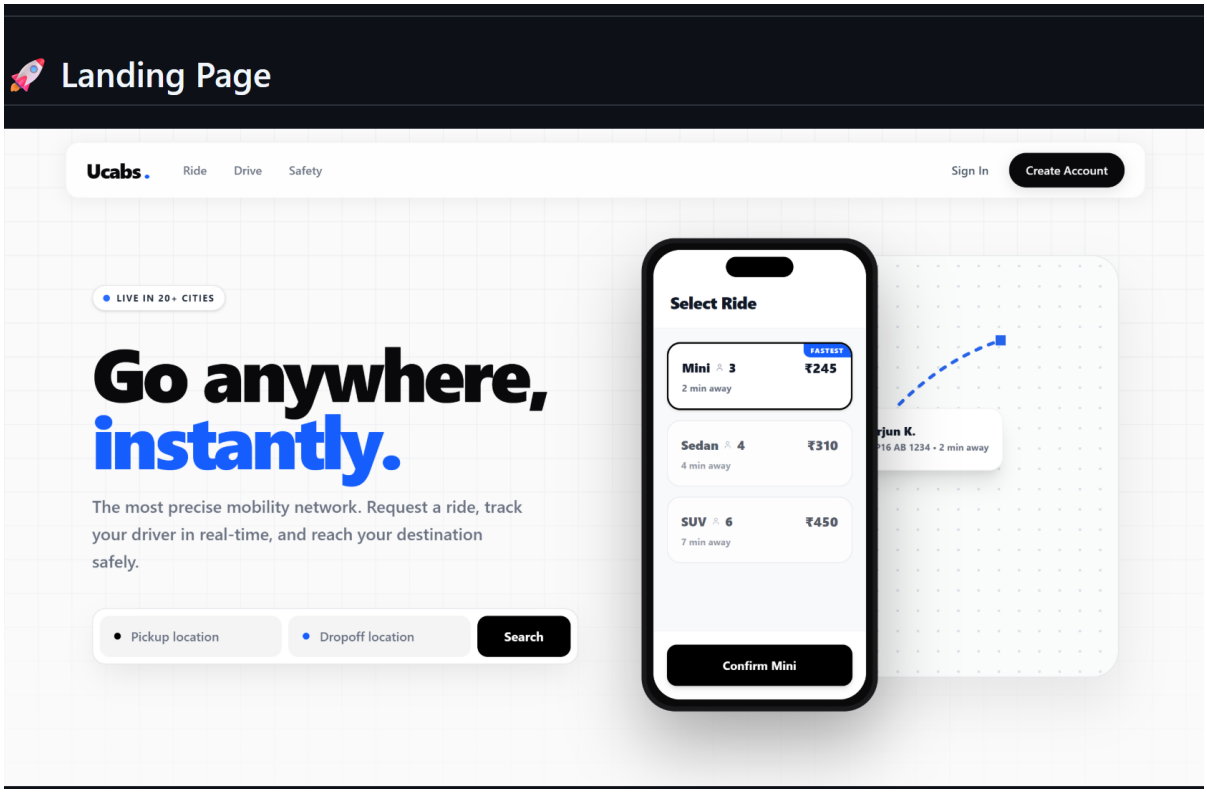
5.1 Project Planning

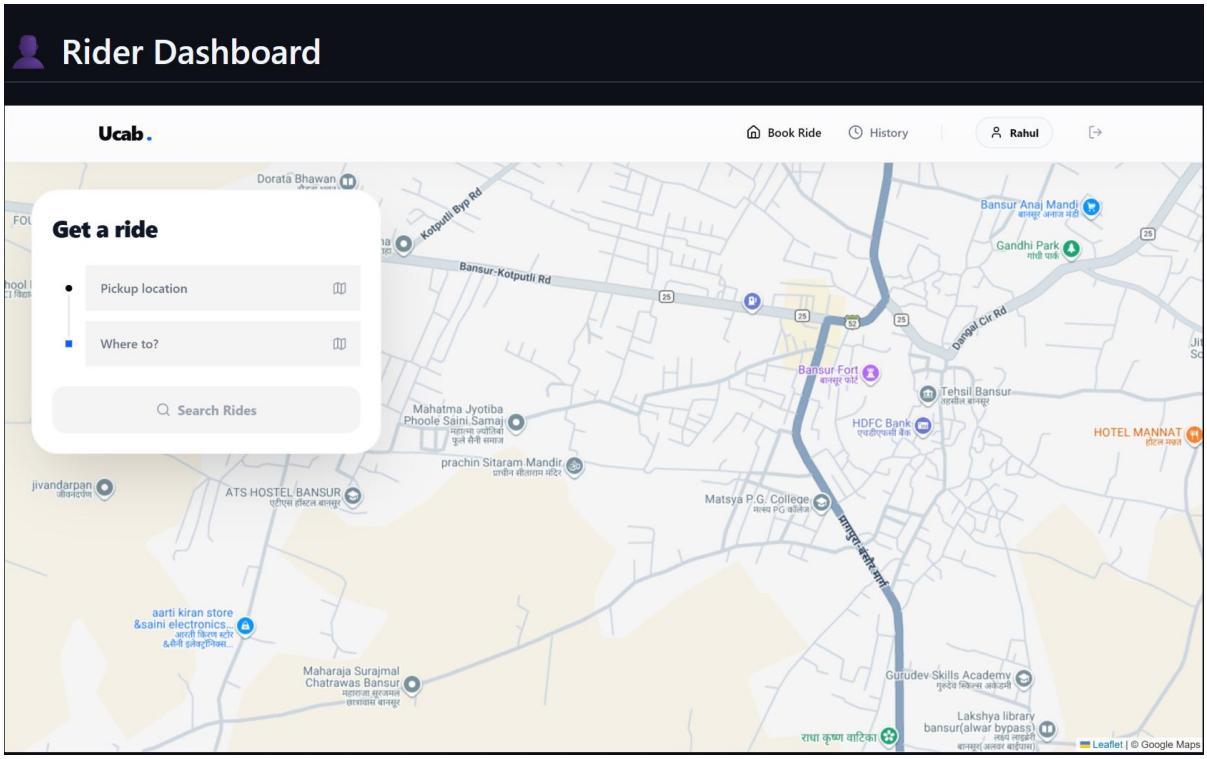
The development process followed the following stages:

1. Requirement Analysis
2. System Design
3. Database Design
4. Frontend Development
5. Backend Development
6. API Integration
7. Testing
8. Deployment

6. RESULTS

6.1 Output Screen shots





7. ADVANTAGES & DISADVANTAGES

Advantages

- **Real-time ride tracking**
- **Secure OTP ride verification**
- **Transparent pricing system**
- **Easy driver management through admin panel**
- **User-friendly interface**

Disadvantages

- **Requires stable internet connection**
- **GPS accuracy depends on device location services**
- **Initial driver network may be limited**

8. CONCLUSION

UCABS successfully demonstrates a modern ride-hailing system using the MERN stack. The platform allows users to book rides easily while enabling drivers to manage trips efficiently.

The integration of real-time communication, secure authentication, and mapping services ensures a reliable and scalable transportation solution.

9. FUTURE SCOPE

Future improvements can include:

- **Online payment integration**

- **Mobile application version**
- **AI-based driver allocation**
- **Push notifications for ride updates**
- **Ride sharing feature**

10. APPENDIX

Source Code

Available on GitHub repository.

Dataset Link

Not applicable (system-based project).

GitHub & Project Demo Link

GitHub Repository: https://github.com/tauqeeralam11/UCabs/